

Generation Interconnection Request System Impact Study Report

For Queue Project Q19-1227

Arcogen 138 kV
July 26, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q19-1227. The Interconnection Customer proposes a 158.0 MW wind facility interconnecting at the Arcogen 138 kV substation in the Virginia area. The request was received on June 15, 2019. The study evaluated system intact and single contingency conditions using the regional planning case, including 450 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$12,381,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Southtown – Commonwealth Chesapeake 345 kV Line, the Lind – Belle Haven Diesel Generation Facility 138 kV loads to 154.9% of its 173 MVA rating. Project DFAX: 8.5%.

The Tasley – Perdue Diesel Generation Facility 138 kV L is overloaded to 149.2% of its 246 MVA emergency rating for the loss of the Southtown – Commonwealth Chesapeake Substation 345 kV. The project contributes 11.3% DFAX (17.8 MW) to this overload.

The Dell Crk – Churchs Ferry 138 kV is overloaded to 137.5% of its 142 MVA emergency rating for the loss of the Cady – Dodd Rd 345 kV. The project contributes 6.0% DFAX (9.5 MW) to this overload.

The Key Largo Substation – Saltspring Island 138 kV is overloaded to 126.3% of its 340 MVA emergency rating for the loss of the Cady – Easton 2 345 kV. The project contributes 11.8% DFAX (18.6 MW) to this overload.

The Saltspring Island – Hillsboro 138 kV is overloaded to 125.6% of its 346 MVA emergency rating for the loss of the Cady – Easton 2 345 kV. The project contributes 11.8% DFAX (18.6 MW) to this overload.

Hansford – Saddlebrooke Ranch 138 kV: 125.1% loading (376 MVA) following the outage of the Southtown – Vienna Operations 345 kV; DFAX = 7.2%.

The Edgewater Station – Whitmore 138 kV Line is overloaded to 124.1% of its 300 MVA emergency rating for the loss of the Cady – Dodd Rd 345 kV. The project contributes 9.4% DFAX (14.8 MW) to this overload.

The Calvert Cliffs Substation 345 kV/138 kV Transformer Bank is overloaded to 123.5% of its 575 MVA emergency rating for the loss of the Easton 2 – Cady 345 kV. The project contributes 24.2% DFAX (38.3 MW) to this overload.

The Bristol – Limon 138 kV Line is overloaded to 122.6% of its 437 MVA emergency rating for the loss of the Cady – Easton 2 345 kV. The project contributes 11.5% DFAX (18.2 MW) to this overload.

Bond-Elk Grove Florin – Gilroy Food 138 kV: 122.4% loading (227 MVA) following the outage of the Vienna Operations – Easton 2 345 kV; DFAX = 7.9%.

The Pit 6 Substation – Limon 138 kV is overloaded to 121.2% of its 257 MVA emergency rating for the loss of the Easton 2 – Vienna Operations 345 kV L. The project contributes 10.2% DFAX (16.1 MW) to this overload.

Henry St Wps – Quinn 138 kV: 120.8% loading (832 MVA) following the outage of the Easton 2 Substation – Vienna Operations 345 kV; DFAX = 25.0%.

For the loss of Cady – Easton 2 345 kV, the Ninemile Pt – Clatskanie 138 kV loads to 120.3% of its 215 MVA rating. Project DFAX: 6.8%.

The Vienna Operations 345 kV/1 kV XFMR is overloaded to 120.0% of its 1,445 MVA emergency rating for the loss of the Vienna Operations – Easton 2 345 kV line. The project contributes 49.4% DFAX (78.1 MW) to this overload.

For the loss of Vienna Operations – Easton 2 345 kV, the Cannelton Hydroelectric Plant – Henry St Wps 138 kV line loads to 119.9% of its 857 MVA rating. Project DFAX: 25.0%.

The Vienna Operations – Easton 2 345 kV is overloaded to 118.8% of its 2,513 MVA emergency rating for the loss of the Ripple – Linde 138 kV. The project contributes 77.5% DFAX (122.5 MW) to this overload.

The Cannelton Hydroelectric Plant – Equus Freeprt Power 138 kV is overloaded to 118.6% of its 892 MVA emergency rating for the loss of the Vienna Operations – Easton 2 Substation 345 kV Line. The project contributes 25.0% DFAX (39.6 MW) to this overload.

For the loss of Easton 2 – Vienna Operations 345 kV, the Yanke – Park Ave 138 kV Line loads to 118.4% of its 908 MVA rating. Project DFAX: 27.1%.

For the loss of Easton 2 Substation – Vienna Operations 345 kV, the Jarratt – Yanke 138 kV loads to 118.2% of its 915 MVA rating. Project DFAX: 27.1%.

For the loss of Vienna Operations – Southtown 345 kV, the Sunst – St Bonifacius Substation 138 kV Line loads to 118.0% of its 1,979 MVA rating. Project DFAX: 54.4%.

Knight St – Equus Freeport Power 138 kV: 117.9% loading (913 MVA) following the outage of the Vienna Operations – Easton 2 345 kV; DFAX = 25.0%.

The Vienna Operations 138/1kV TX is overloaded to 117.7% of its 1,472 MVA emergency rating for the loss of the Easton 2 – Vienna Operatins 345 kV. The project contributes 49.4% DFAX (78.1 MW) to this overload.

For the loss of Vienna Operations – Easton 2 345 kV, the Shoals – Bristol 138 kV loads to 117.4% of its 311 MVA rating. Project DFAX: 7.8%.

Henning – Kiwi 138 kV: 117.3% loading (1,068 MVA) following the outage of the Vienna Operations – Easton 2 Substation 345 kV Line; DFAX = 32.9%.

Knight Street – Long Lake 138 kV: 116.7% loading (949 MVA) following the outage of the Vienna Operations – Easton 2 345 kV line; DFAX = 25.0%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Lind – Belle Haven Diesel Generation Facility 138 kV	Rebuild line with higher-capacity conductor	\$11,053,000	6% (\$678,000)
Tasley – Perdue Diesel Generation Facility 138 kV L	Rebuild line with higher-capacity conductor	\$4,016,000	5% (\$198,000)
Dell Crk – Churchs Ferry 138 kV	Rebuild line with higher-capacity conductor	\$2,487,000	3% (\$79,000)
Saltspring Island – Key Largo 138 kV line	Rebuild line with higher-capacity conductor	\$24,891,000	3% (\$804,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Saltspring Island – Hillsboro 138 kV Line	Rebuild line with higher-capacity conductor	\$7,717,000	3% (\$249,000)
Hansford – Saddlebrooke Ranch 138 kV Line	Rebuild line with higher-capacity conductor	\$8,605,000	3% (\$286,000)
Edgewater Station – Whitmore 138 kV Line	Rebuild line with higher-capacity conductor	\$7,087,000	3% (\$226,000)
Calvert Cliffs Substation 345 kV/138 kV Transformer Bank	Replace transformer with higher-rated unit	\$11,412,000	3% (\$366,000)
Bristol – Limon 138 kV L	Rebuild line with higher-capacity conductor	\$3,765,000	3% (\$120,000)
Bond-Elk Grove Florin Substation – Gilroy Food 138 kV	Rebuild line with higher-capacity conductor	\$44,116,000	3% (\$1,505,000)
Pit 6 – Limon 138 kV L	Rebuild line with higher-capacity conductor	\$15,327,000	3% (\$510,000)
Quinn – Henry St Wps 138 kV	Rebuild line with higher-capacity conductor	\$2,213,000	4% (\$81,000)
Ninemile Pt – Clatskanie 138 kV	Rebuild line with higher-capacity conductor	\$36,576,000	3% (\$1,154,000)
Vienna Operations 345 kV/1 kV XFMR	Replace transformer with higher-rated unit	\$17,687,000	4% (\$705,000)
Cannelton Hydroelectric Plant – Henry St Wps 138 kV line	Rebuild line with higher-capacity conductor	\$5,288,000	4% (\$193,000)
Vienna Operations – Easton 2 345 kV	Rebuild line with higher-capacity conductor	\$80,744,000	4% (\$2,954,000)
Cannelton Hydroelectric Plant – Equus Freeport Power 138 kV	Rebuild line with higher-capacity conductor	\$3,439,000	4% (\$125,000)
Yanke – Park Ave 138 kV Line	Rebuild line with higher-capacity conductor	\$7,644,000	4% (\$275,000)
Jarratt – Yanke 138 kV	Rebuild line with higher-capacity conductor	\$10,184,000	4% (\$367,000)
St Bonifacius – Sunset 138 kV	Rebuild line with higher-capacity conductor	\$6,462,000	3% (\$221,000)
Knight Steet – Equus Freeport Power 138 kV	Rebuild line with higher-capacity conductor	\$2,008,000	4% (\$73,000)
Vienna Operations 138/1kV TX	Replace transformer with higher-rated unit	\$8,949,000	4% (\$357,000)
Shoals – Bristol 138 kV	Rebuild line with higher-capacity conductor	\$9,793,000	4% (\$341,000)
Henning – Kiwi 138 kV	Rebuild line with higher-capacity conductor	\$2,676,000	4% (\$100,000)
Knight Street – Long Lake 138 kV	Rebuild line with higher-capacity conductor	\$11,347,000	4% (\$414,000)
	Total allocated to project		\$12,381,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.